Inference for numerical data

Loading packages

```
library(tidyverse)
library(openintro)
library(infer)
```

```
data('yrbss', package='openintro')

?yrbss
```

```
starting httpd help server ... done
```

```
glimpse(yrbss)
```

```
Rows: 13,583
Columns: 13
$ age                      <int> 14, 14, 15, 15, 15, 15, 15, 14, 15, 15, 15, 1~
$ gender                   <chr> "female", "female", "female", "female", "fema~
$ grade                    <chr> "9", "9", "9", "9", "9", "9", "9", "9", "9", ~
$ hispanic                 <chr> "not", "not", "hispanic", "not", "not", "not"~
$ race                     <chr> "Black or African American", "Black or Africa~
$ height                   <dbl> NA, NA, 1.73, 1.60, 1.50, 1.57, 1.65, 1.88, 1~
$ weight                   <dbl> NA, NA, 84.37, 55.79, 46.72, 67.13, 131.54, 7~
$ helmet_12m               <chr> "never", "never", "never", "never", "did not ~
$ text_while_driving_30d   <chr> "0", NA, "30", "0", "did not drive", "did not~
$ physically_active_7d     <int> 4, 2, 7, 0, 2, 1, 4, 4, 5, 0, 0, 0, 4, 7, 7, ~
$ hours_tv_per_school_day  <chr> "5+", "5+", "5+", "2", "3", "5+", "5+", "5+",~
$ strength_training_7d     <int> 0, 0, 0, 0, 1, 0, 2, 0, 3, 0, 3, 0, 0, 7, 7, ~
$ school_night_hours_sleep <chr> "8", "6", "<5", "6", "9", "8", "9", "6", "<5"~
```

Exercise 1

There are a total of 13,583 cases in this dataset and each is a high schooler in 9th through 12th grade

```
summary(yrbss$weight)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
  29.94   56.25   64.41   67.91   76.20  180.99    1004
```

Exercise 2

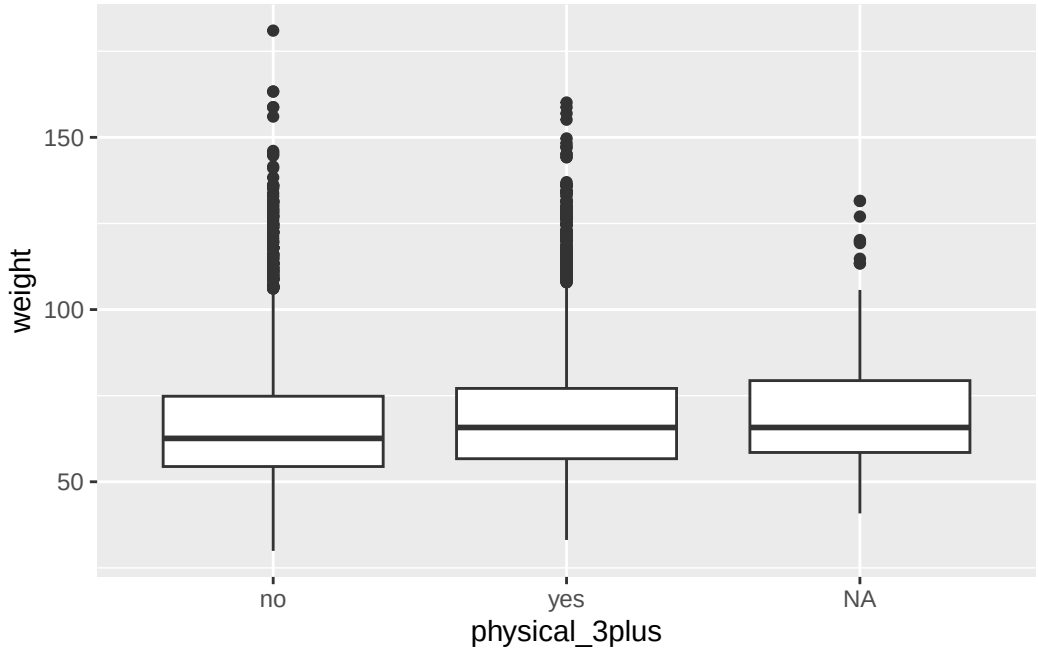
We are missing a total of 1004 entries

```
yrbss <- yrbss %>%
  mutate(physical_3plus = ifelse(yrbss$physically_active_7d > 2, "yes", "no"))
```

Exercise 3

```
ggplot(yrbss, aes(x = physical_3plus, y = weight)) +
  geom_boxplot()
```

```
Warning: Removed 1004 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



The boxplot shows that the weights of the more active highschool kids is ever so slightly higher than the non-active ones. However, the data is very close.

```
yrbss %>%
  group_by(physical_3plus) %>%
  summarise(mean_weight = mean(weight, na.rm = TRUE))
```

```
# A tibble: 3 x 2
  physical_3plus mean_weight
  <chr>                <dbl>
1 no                    66.7
2 yes                   68.4
3 <NA>                  69.9
```

Exercise 4

The conditions for inference should be satisfied. The data comes from a simple random source, they should all be independent from one another and the data source is sufficiently large to create a normal distribution when sampling means.

Exercise 5

H_0 : Being active does not affect weight
H_a : Being active does affect weight

```
obs_diff <- yrbss %>%
  drop_na(physical_3plus) %>%
  specify(weight ~ physical_3plus) %>%
  calculate(stat = "diff in means", order = c("yes", "no"))
```

```
Warning: Removed 946 rows containing missing values.
```

```
null_dist <- yrbss %>%
  drop_na(physical_3plus) %>%
  specify(weight ~ physical_3plus) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("yes", "no"))
```

```
Warning: Removed 946 rows containing missing values.
```

```
print(obs_diff)
```

```
Response: weight (numeric)
Explanatory: physical_3plus (factor)
# A tibble: 1 x 1
   stat
  <dbl>
1  1.77
```

```
print(null_dist)
```

```
Response: weight (numeric)
Explanatory: physical_3plus (factor)
Null Hypothes...
# A tibble: 1,000 x 2
   replicate    stat
       <int>   <dbl>
 1         1  0.266
```

```
 2         2  0.538
 3         3  0.431
 4         4 -0.122
 5         5 -0.482
 6         6  0.218
 7         7  0.530
 8         8 -0.0231
 9         9 -0.245
10        10  0.485
# i 990 more rows
```

Exercise 6

```
null_dist %>%
  filter(stat >= obs_diff$stat) %>%
  summarise(n_at_least_obs = n())
```

```
# A tibble: 1 x 1
  n_at_least_obs
           <int>
1              0
```

Based on the above, none of our null permutations has a difference of at least obs_stat

```
null_dist %>%
  get_p_value(obs_stat = obs_diff, direction = "two_sided")
```

```
Warning: Please be cautious in reporting a p-value of 0. This result is an approximation
based on the number of `reps` chosen in the `generate()` step.
i See `get_p_value()` (`?infer::get_p_value()`) for more information.
```

```
# A tibble: 1 x 1
  p_value
    <dbl>
1       0
```

Exercise 7

```r
boot_dist <- yrbss %>%
  drop_na(physical_3plus, weight) %>%
  specify(weight ~ physical_3plus) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "diff in means", order = c("yes", "no"))

ci <- boot_dist %>%
  get_confidence_interval(level = 0.95, type = "percentile")

ci
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
     <dbl>    <dbl>
1     1.13     2.46
```

Exercise 8

```r
boot_height <- yrbss %>%
  drop_na(height) %>%
  specify(response = height) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "mean")

ci_height <- boot_height %>%
  get_confidence_interval(level = 0.95, type = "percentile")

ci_height
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
     <dbl>    <dbl>
1     1.69     1.69
```

Exercise 9

```r
ci_height_90 <- boot_height %>%
  get_confidence_interval(level = 0.90, type = "percentile")

ci_height_90
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
     <dbl>    <dbl>
1     1.69     1.69
```

The confidence interval seemed to not move.

Exercise 10

```r
obs_diff_height <- yrbss %>%
  drop_na(physical_3plus, height) %>%
  specify(height ~ physical_3plus) %>%
  calculate(stat = "diff in means", order = c("yes", "no"))

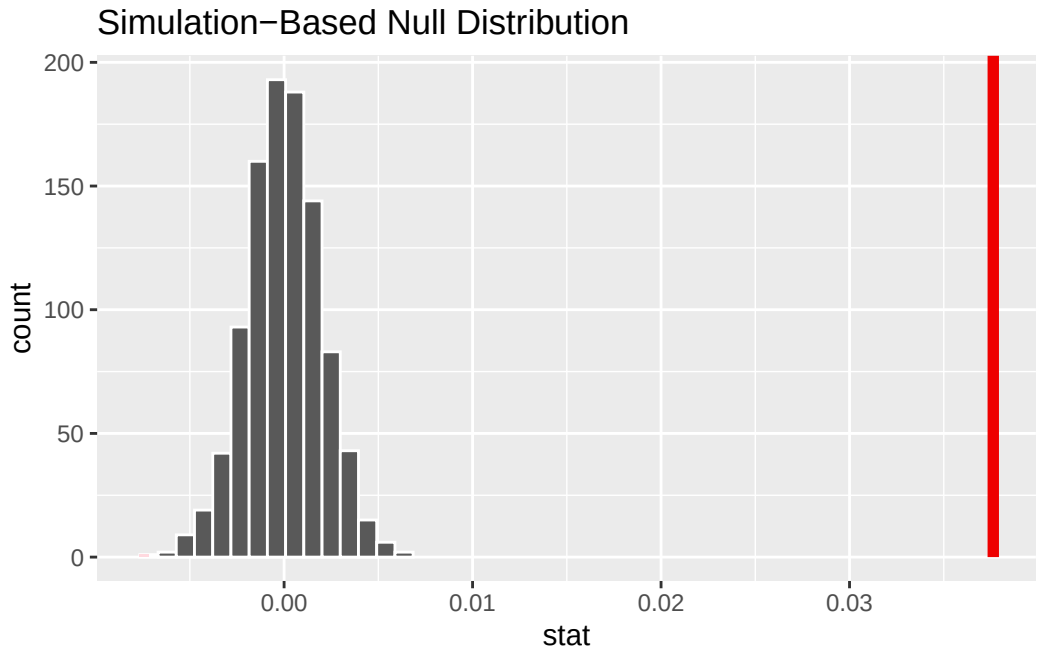
null_dist_height <- yrbss %>%
  drop_na(physical_3plus, height) %>%
  specify(height ~ physical_3plus) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("yes", "no"))

null_dist_height %>%
  get_p_value(obs_stat = obs_diff_height, direction = "two-sided")
```

```
Warning: Please be cautious in reporting a p-value of 0. This result is an approximation
based on the number of `reps` chosen in the `generate()` step.
i See `get_p_value()` (`?infer::get_p_value()`) for more information.
```

```
# A tibble: 1 x 1
  p_value
    <dbl>
1       0
```

```r
visualize(null_dist_height) +
  shade_p_value(obs_stat = obs_diff_height, direction = "two-sided")
```



There seems to be no difference at all when comparing height and activity level.

Exercise 11

```r
yrbss %>%
  count(hours_tv_per_school_day)
```

```
# A tibble: 8 x 2
  hours_tv_per_school_day     n
  <chr>                   <int>
1 1                        1750
2 2                        2705
3 3                        2139
4 4                        1048
5 5+                       1595
6 <1                       2168
7 do not watch             1840
8 <NA>                      338
```

There are a total of 7 options with one extra being an NA value

Exercise 12

We could ask the question "Is there a difference in mean weight between those who get fewer than 7 hours of sleep on school night and those who get more than 8 hours?" we can use $\alpha = .05$

```
yrbss <- yrbss %>%
  mutate(sleep_group = case_when(
    school_night_hours_sleep %in% c("<5", "5", "6") ~ "under7",
    school_night_hours_sleep %in% c("7", "8", "9", "10+") ~ "7plus",
    TRUE ~ NA_character_
  ))

sleep_data <- yrbss %>% drop_na(sleep_group, weight)

sleep_data %>%
  group_by(sleep_group) %>%
  summarise(n = n(), mean_weight = mean(weight), sd_weight = sd(weight))
```

```
# A tibble: 2 x 4
  sleep_group     n mean_weight sd_weight
  <chr>       <int>       <dbl>     <dbl>
1 7plus        6748        67.3      16.4
2 under7       4733        68.7      17.7
```

```
obs_diff_sleep <- sleep_data %>%
  specify(weight ~ sleep_group) %>%
  calculate(stat = "diff in means", order = c("under7", "7plus"))

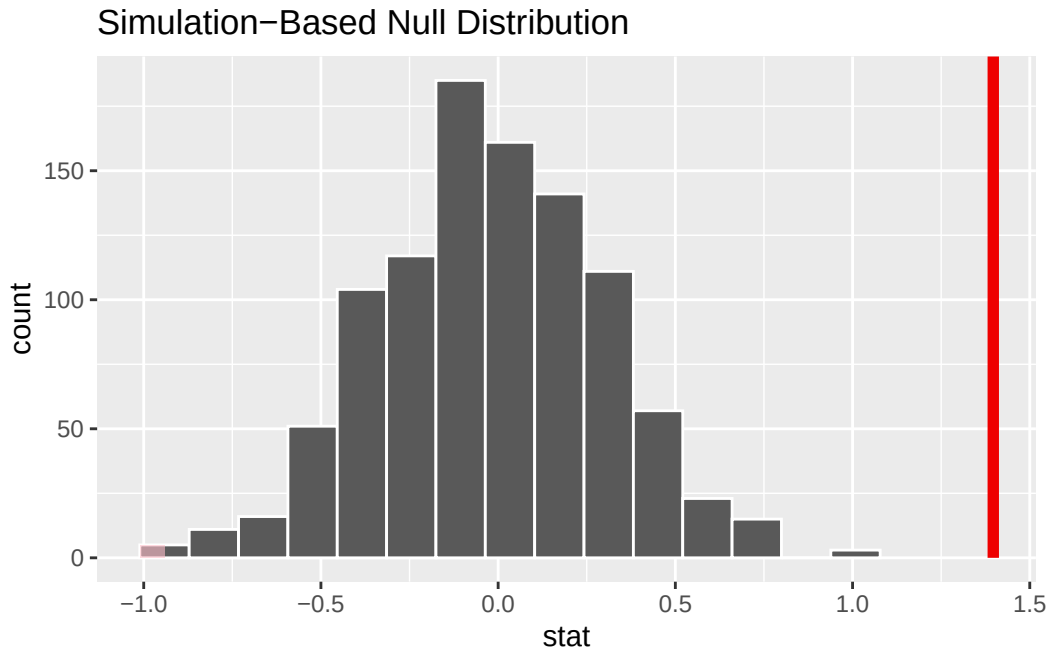
null_dist_sleep <- sleep_data %>%
  specify(weight ~ sleep_group) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("under7", "7plus"))

null_dist_sleep %>%
  get_p_value(obs_stat = obs_diff_sleep, direction = "two-sided")
```

```
Warning: Please be cautious in reporting a p-value of 0. This result is an approximation
based on the number of `reps` chosen in the `generate()` step.
i See `get_p_value()` (`?infer::get_p_value()`) for more information.

# A tibble: 1 x 1
  p_value
    <dbl>
1       0
```

```r
visualize(null_dist_sleep) +
  shade_p_value(obs_stat = obs_diff_sleep, direction = "two-sided")
```




```r
obs_diff_sleep <- sleep_data %>%
  specify(weight ~ sleep_group) %>%
  calculate(stat = "diff in means", order = c("under7", "7plus"))

null_dist_sleep <- sleep_data %>%
  specify(weight ~ sleep_group) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("under7", "7plus"))
```

```r
null_dist_sleep %>%
  get_p_value(obs_stat = obs_diff_sleep, direction = "two-sided")
```

```
Warning: Please be cautious in reporting a p-value of 0. This result is an approximation
based on the number of `reps` chosen in the `generate()` step.
i See `get_p_value()` (`?infer::get_p_value()`) for more information.
```

```
# A tibble: 1 x 1
  p_value
    <dbl>
1       0
```

```r
boot_sleep <- sleep_data %>%
  specify(weight ~ sleep_group) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "diff in means", order = c("under7", "7plus"))

get_confidence_interval(boot_sleep, level = 0.95, type = "percentile")
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
     <dbl>    <dbl>
1    0.736     2.04
```

With a ci as wide as this, we can confidently say that the students who sleep less weigh .748 and 2.0 kg more on average than those who regularly sleep more than 7 hours on school nights.